Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 1 of 10 · Question 5 part b.i · 4 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

5 (a) (i)	Eustachian tube (1)		1
(ii)	any three from • pinna (1) • semi-circular canals (1) • auditory nerve (1) • cochlea (1)	Ignore oval/round window/ultriculus / stacculus Allow organ of corti	3
(b) (i)	4 of • transmit vibration (1) • from ear drum/tympanum (1) • across middle ear/to (named) ossicles (1) • to oval window/cochlea (1) • amplify vibration (1)		4 max
(ii)	A; (air)		1
(c)	• equalises pressure (1) • either side of eardrum (1) • which allows it to vibrate/transmit vibrations (1)	Allow tympanic membrane	3

Total 12 marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 1 of 10 · Question 5 part b.i · 4 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 2 of 10 · Question 5 part a.i · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

5 (a) (i)	Eustachian tube (1)		1
(ii)	any three from • pinna (1) • semi-circular canals (1) • auditory nerve (1) • cochlea (1)	Ignore oval/round window/ultriculus / stacculus Allow organ of corti	3
(b) (i)	4 of • transmit vibration (1) • from ear drum/tympanum (1) • across middle ear/to (named) ossicles (1) • to oval window/cochlea (1) • amplify vibration (1)		4 max
(ii)	A; (air)		1
(c)	• equalises pressure (1) • either side of eardrum (1) • which allows it to vibrate/transmit vibrations (1)	Allow tympanic membrane	3

Total 12 marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 2 of 10 · Question 5 part a.i · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 3 of 10 · Question 5 part a.ii · 3 marks

2020 NOV · Paper 02 · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 4 of 10 · Question 5 part b.ii · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

5 (a) (i)	Eustachian tube (1)		1
(ii)	any three from • pinna (1) • semi-circular canals (1) • auditory nerve (1) • cochlea (1)	Ignore oval/round window/ultriculus / stacculus Allow organ of corti	3
(b) (i)	4 of • transmit vibration (1) • from ear drum/tympanum (1) • across middle ear/to (named) ossicles (1) • to oval window/cochlea (1) • amplify vibration (1)		4 max
(ii)	A; (air)		1
(c)	• equalises pressure (1) • either side of eardrum (1) • which allows it to vibrate/transmit vibrations (1)	Allow tympanic membrane	3

Total 12 marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 4 of 10 · Question 5 part b.ii · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 5 of 10 · Question 5 part a.i · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

5 (a) (i)	A = eardrum; B = cochlea;		2
(ii)	Any two from the following • collects sound waves/vibrations/determines direction of sound; • directs them through the auditory / ear canal; • to the eardrum;		2
(b) (i)	• axes labels; • units; • bars correctly plotted; • scale;		4
(ii)	• (20 – 14) 6; • (6 ÷20) -0.30; • (x100) -30.00%	Allow ecf is minus symbol not given (Max 2 marks)	3
(c) (i)	Any two from the following • gender variation; • some people may have an ear defect;		2
(ii)	EITHER • check results; • with (data) from another group; OR		2

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 5 of 10 · Question 5 part a.i · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

	• repeat the experiment; • calculate mean;		
			Total 15

Question number	Answer	Notes	Marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 6 of 10 · Question 5 part a.ii · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

5 (a) (i)	A = eardrum; B = cochlea;		2
(ii)	Any two from the following • collects sound waves/vibrations/determines direction of sound; • directs them through the auditory / ear canal; • to the eardrum;		2
(b) (i)	• axes labels; • units; • bars correctly plotted; • scale;		4
(ii)	• (20 – 14) 6; • (6 ÷20) -0.30; • (x100) -30.00%	Allow ecf is minus symbol not given (Max 2 marks)	3
(c) (i)	Any two from the following • gender variation; • some people may have an ear defect;		2
(ii)	EITHER • check results; • with (data) from another group; OR		2

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 6 of 10 · Question 5 part a.ii · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

	• repeat the experiment; • calculate mean;		
			Total 15

Question number	Answer	Notes	Marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 7 of 10 · Question 5 part b.iii · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number					
5	(a)	(i)	• use people from a range of ages; • hearing test using high frequency sounds; • control other variables e.g. sex, other health issues/same distance from sound source;		3
		(ii)	exposure to loud noises/genetic cause/infection		1
	(b)	(i)	• vibrations of the eardrum not transferred to the ossicles; • vibrations are not transferred to the cochlea; • hair cells are not stimulated/impulses not generated/not transferred to the auditory nerve/brain;		3
		(ii)	Two from: • conditions are moist/warm; • ideal conditions for bacterial growth/to multiply; • pathogens/bacteria more likely to get trapped;		2
		(iii)	Three from: • semi-circular canals; • contain fluid (in three tubes); • as the head moves so does the fluid; • transmits the information to the brain/cerebellum; • (cerebellum)co-ordinates a response to maintain balance/control balance;		3

Total 12 marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 7 of 10 · Question 5 part b.iii · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 8 of 10 · Question 5 part c · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

5 (a) (i)	Eustachian tube (1)		1
(ii)	any three from • pinna (1) • semi-circular canals (1) • auditory nerve (1) • cochlea (1)	Ignore oval/round window/ultriculus / stacculus Allow organ of corti	3
(b) (i)	4 of • transmit vibration (1) • from ear drum/tympanum (1) • across middle ear/to (named) ossicles (1) • to oval window/cochlea (1) • amplify vibration (1)		4 max
(ii)	A; (air)		1
(c)	• equalises pressure (1) • either side of eardrum (1) • which allows it to vibrate/transmit vibrations (1)	Allow tympanic membrane	3

Total 12 marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 8 of 10 · Question 5 part c · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 9 of 10 · Question 5 part b.i · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
5	(a)	(i)	• use people from a range of ages; • hearing test using high frequency sounds; • control other variables e.g. sex, other health issues/same distance from sound source;		3
		(ii)	exposure to loud noises/genetic cause/infection		1
	(b)	(i)	• vibrations of the eardrum not transferred to the ossicles; • vibrations are not transferred to the cochlea; • hair cells are not stimulated/impulses not generated/not transferred to the auditory nerve/brain;		3
		(ii)	Two from: • conditions are moist/warm; • ideal conditions for bacterial growth/to multiply; • pathogens/bacteria more likely to get trapped;		2
		(iii)	Three from: • semi-circular canals; • contain fluid (in three tubes); • as the head moves so does the fluid; • transmits the information to the brain/cerebellum; • (cerebellum)co-ordinates a response to maintain balance/control balance;		3

Total 12 marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 9 of 10 · Question 5 part b.i · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 10 of 10 · Question 5 part a.ii · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
5	(a)	(i)	• use people from a range of ages; • hearing test using high frequency sounds; • control other variables e.g. sex, other health issues/same distance from sound source;		3
		(ii)	exposure to loud noises/genetic cause/infection		1
	(b)	(i)	• vibrations of the eardrum not transferred to the ossicles; • vibrations are not transferred to the cochlea; • hair cells are not stimulated/impulses not generated/not transferred to the auditory nerve/brain;		3
		(ii)	Two from: • conditions are moist/warm; • ideal conditions for bacterial growth/to multiply; • pathogens/bacteria more likely to get trapped;		2
		(iii)	Three from: • semi-circular canals; • contain fluid (in three tubes); • as the head moves so does the fluid; • transmits the information to the brain/cerebellum; • (cerebellum)co-ordinates a response to maintain balance/control balance;		3

Total 12 marks

Type MS · Identify ear parts and use the sound path to explain hearing and hearing loss

Answer 10 of 10 · Question 5 part a.ii · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks